Evaporation and crystallisation

Filtration cannot be used to separate substances that have been dissolved in a solvent, because the particles are too small and pass through the filter paper. Heat can be used to separate a liquid from a solution, leaving the solute behind. This process is called **evaporation**.

☞ **1** Circle the correct option from each pair in brackets.

Filtration is used to remove (soluble/insoluble) substances from a liquid.

Evaporation is used to remove (soluble/insoluble) substances from a liquid.

When you separate a solution using evaporation, you (keep/lose) the solvent.

When you separate a solution using evaporation, you are left with the (solute/solvent).

Evaporating the water from a solution by heating is a quick method of separating the solute from the solvent. However, sometimes evaporation is not a suitable method because the quantities are too large, or you want large crystals. In these cases, **crystallisation** can be used instead. For example, in the production of commercial salt, the sun's heat is used to evaporate the water. Large salt crystals form because the water is evaporated slowly.

2 Decide whether each of the following statements is true or false.

Evaporation allows you to keep the separated solvent and solute. _____

Filtration can't be used to separate the solute and solvent from a solution. _____

If you evaporate the liquid quickly, the crystals formed are small. _____

Crystallisation occurs when a solution is heated quickly. _____

Filtration is used to remove insoluble substances from a liquid. _____

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